

Land Use Change and Avian Disease Dynamics

Maureen H. Murray and Sonia M. Hernandez

9.1 Introduction

Birds live on a human-dominated planet. Over 75% of the Earth's ice-free land has been disturbed by human land uses (Ellis and Ramankutty 2008), impacting ecosystems around the globe. Disturbances associated with human activities such as land clearing, fragmentation, and pollution influence the relationships between birds, their pathogens, and the environment they share. Such shifts in disease dynamics can arise through the impacts of land use change on aspects of hosts, vectors, and/or pathogens, including vector and host abundance, behavior, and physiology, and through pathogen persistence in the environment. To address this complexity, we first describe the major causes of anthropogenic land use change that can impact birds and their pathogens across diverse ecosystems. We next discuss key changes associated with land use, such as habitat loss or fragmentation, pollution, and anthropogenic resources, that are relevant to avian disease ecology. We then synthesize documented associations between avian health and urbanization, the fastest growing type of land use change on Earth (Antrop and Van Eetvelde 2000). We close this chapter with a consideration of land use changes in a One Health context and future directions for advancing avian disease ecology in rapidly changing landscapes.

9.1.1 Major drivers of land use change

Global declines of many avian species can be attributed to habitat loss and degradation, primarily

from urbanization, deforestation, and agriculture (Rosenberg et al. 2019). Urban and agricultural areas are typically associated with deforestation to create cropland or built environments, although logging and other extractive industries contribute to deforestation globally. Agriculture is the most widespread land use on Earth with approximately 51 million km² of land—50% of Earth's habitable land area—used for cropland or pasture (Ritchie and Roser 2020). To accommodate global demand for crops, land clearing for agriculture typically involves clearing forest or other native vegetation through logging, burning, or draining wetlands, managing soil through tilling and fertilization, and maintaining plant monocultures and/or production animals. Based on its extent and disturbance, agricultural intensification is considered the largest driver of biodiversity loss globally (Dudley and Alexander 2017) and avian species that inhabit agricultural areas are declining faster than birds associated with any other biome (Murphy 2003).

Although urban areas only account for a small proportion of Earth's habitable land area, they are the fastest growing habitat type on Earth due to a rapidly urbanizing human population (Venter et al. 2016). With urban development comes profound changes in abiotic and biotic conditions that can impact the ecology of avian hosts, vectors, and pathogens (reviewed in Bradley and Altizer 2007). Increased impervious surfaces associated with roads and buildings absorb solar radiation and lead to higher temperatures in urban centers than the surrounding countryside, a phenomenon known as the heat island effect (Oke 1973). Increasing

temperatures could impact diverse components of disease dynamics including the behavior or physiology of hosts and vectors and/or pathogen persistence in the environment. Urban areas are also more likely to contain non-native species than surrounding landscapes from either intentional or unintentional human introductions (Gaertner et al. 2017), leading to altered host and vector communities.

The global loss of natural wetlands has forced aquatic birds to become increasingly dependent on alternative and artificial habitats, but the use of such wetlands by waterbirds is considered opportunistic, and little consideration has been given for the potential implications to pathogen dynamics in these environments. Sewage treatment plants with waste ponds and wastewater treatment plants have been documented as significant water-bird habitat (Murray et al. 2014). However, these modified wetland environments can be heavily polluted (Hamilton 2007) and may alter pathogen dynamics by facilitating environmental persistence of pathogens, increased production of toxins, or antimicrobial resistance.

9.2 Key changes associated with land use and influences on disease dynamics

Anthropogenic land use, regardless of cause, is typically accompanied by a set of key changes that can impact disease dynamics for birds (Figure 9.1). Land conversion for the purposes of urbanization, cropland, or resource extraction typically causes habitat loss (e.g., removal of a forest stand), habitat degradation (i.e., a reduction in quality), and fragmentation (e.g., bisecting habitat patches with roads or other linear features). These processes are also typically accompanied by increases in human disturbance, which can generate pollution from industrial processes and anthropogenic resources provided intentionally (e.g., bird feeding) or unintentionally (e.g., landfills) to birds. These landscape changes can have several effects on avian hosts, vectors, and pathogens relevant to disease transmission. Altered habitat conditions can lead to changes in the abundance, community composition, or behavior of hosts and vectors, ultimately changing the likelihood of contact with other hosts or vectors, and thus pathogen transmission

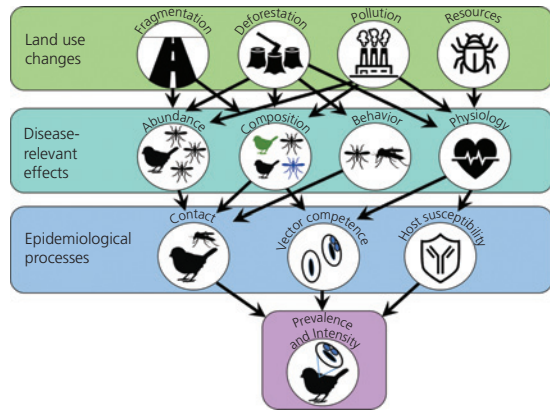


Figure 9.1 A conceptual framework using the example of avian malaria (caused by *Plasmodium* spp.) to illustrate how prevalence and intensity of a vector-borne pathogen (purple) can be influenced by some of the key changes associated with land use (green bar), their downstream effects on individuals and populations (teal bar), and associated consequences for several epidemiological processes (blue bar). Here we conceptualize how habitat degradation can alter abundance and community composition of avian hosts and mosquito vectors, biting behavior of mosquitoes, and physiology of hosts and vectors in ways that alter their susceptibility and competence, respectively. While we selectively show the arrows likely to be most important for avian malaria, changes to 'Resources' from human land use can also alter host and vector abundance, community composition, and behaviors in ways relevant for many other avian–pathogen systems. (Icons were created by Norbert Kucsera, Audrey Laymik, Iconographer, Bohdan Burmich, Symbolon, Alex Tai of Noun Project.)

(Figure 9.1). Habitat changes can also impact host and vector physiology, potentially increasing susceptibility to infection or vector competence. Finally, while not illustrated in our conceptual model of a vector-borne pathogen, land use changes can alter the ability of a parasite or pathogen to persist in the external environment. Because of the myriad potential effects on avian disease ecology, we will discuss each key change and their respective impacts on hosts, vectors, and pathogens in turn.

9.2.1 Habitat loss, degradation, and fragmentation

The primary mechanisms by which habitat loss, degradation, and fragmentation can alter disease dynamics and risk include changes in opportunities for contact among hosts or between hosts and vectors and changes in host and vector competence/susceptibility due to environmental stress. It can be

challenging to interpret and predict the individual effects of habitat loss, degradation, and fragmentation on avian disease dynamics because fragmentation is nearly always accompanied by habitat loss (Fahrig 2003). For example, although fragmentation primarily alters opportunities for contact between host and vector species by introducing edges between habitat types, fragmentation is often associated with habitat degradation due to edge effects and the human disturbance that accompanies development. The term habitat loss is also species-specific; while fragmentation reduces the habitat available to interior species (e.g., northern spotted owls, *Strix occidentalis caurina*), it creates more habitat for edge-loving species (e.g., common blackbirds, *Turdus merula*). These factors complicate our understanding of the underlying mechanisms driving changes in avian disease dynamics with land use and emphasize that a synthetic view is necessary.

One of the primary ways that habitat loss and degradation alter disease dynamics is via changes to host and/or vector abundance and thus host and vector community composition. Bird abundance typically declines in degraded landscapes due to lack of habitat-associated resources and, perhaps to a lesser extent, habitat connectivity. For example, a loss of forest cover from 95% to 50% was associated with the extirpation of 33 forest bird species in the Colombian Andes (Palacio et al. 2019). A reduction in host density likely reduces opportunities for pathogens that have density-dependent transmission (see Chapter 7). However, extinction risk from habitat loss is higher for habitat specialists (Owens and Bennett 2000) and disturbed landscapes are often colonized by non-native species such as house sparrows (*Passer domesticus*) and rock doves (*Columba livia*). These changes in host community composition and novel opportunities for contact with non-native species can lead to enhanced exposure to pathogens. For example, urban raptors have a higher infection prevalence of *Trichomonas gallinae*, the causative agent of trichomoniasis, than their non-urban counterparts because urban raptors prey on higher proportions of Columbiformes that are significant reservoirs (Boal et al. 1998; see also Chapter 8). Thus, although habitat loss may reduce the likelihood of conspecific contact by reducing

host abundance (but see barriers to movement below), it can sometimes increase pressure from novel host contact.

Habitat loss and fragmentation can also create opportunities for contact between birds and insect vectors (Figure 9.1). These opportunities could arise because of shifts in host communities or microclimate conditions suitable for vectors. For example, the proportion of birds with attached ticks in Brazil's Atlantic forest was higher in smaller, degraded forest patches where bird species diversity was lower (Ogrzewalska et al. 2011). Degraded patches were dominated by relatively few generalist bird species, potentially promoting vector transmission; however, the environmental conditions in these patches may also be favorable for free-living tick life stages (Ogrzewalska et al. 2011). At a microhabitat scale, the removal of canopy cover typically associated with land clearing and fragmentation could impact the diversity and abundance of insect vectors through the availability of direct sunlight, humidity, edge habitat, soil structure, and standing water. For example, in Southeast Asia, survival of *Anopheles* mosquitoes was higher in deforested areas, where ambient temperatures were higher (Zhong et al. 2016). Relevant to these rapid changes in microhabitat for insect vectors, a growing body of work has examined changes in avian blood parasites in birds with deforestation (Sehgal 2010; Sehgal 2015). The prevalence of avian malaria can often be relatively higher in disturbed forests, likely due to an increase in the range of the mosquito vector. For example, deforestation in Alaka'i Wilderness Preserve on Kaua'i Island in Hawai'i may shift seasonal pulses in malaria prevalence for Hawaiian honeycreepers to a year-round risk of infection by increasing the altitudinal range of vector populations (Atkinson and LaPointe 2009). However, the relationship between deforestation and blood parasites can be complex, with some *Plasmodium* morphospecies increasing and others decreasing in disturbed forests (Chasar et al. 2009). Thus, the interactions between hosts and vectors in disturbed forests are complicated by species-specific differences in mosquito responses to forest disturbance in interior, edge, and open habitats.

The configuration of disturbed and conservation areas may have important consequences for wildlife

persistence and pathogen transmission in areas with dynamic land use change. For example, although more integration between agricultural and natural areas (a 'land sharing' system) may lead to proportionately more fragmented habitat patches, model simulations demonstrate that such an approach may mitigate risk of avian malaria relative to large-scale nature reserves (a 'land sparing' approach) because small remnant forest patches are important in reducing malaria prevalence in surrounding non-forested habitats (Mendenhall et al. 2013). Dipteran vectors also respond to the configuration of forest management, with haemosporidian infections in Diptera insects increasing with more edge habitat, lower shrub density, less leaf litter, and lower humidity, all characteristics of harvested forest stands (van Hoesel et al. 2019). Overall, anticipating changes in vector-borne pathogens for birds with land use change will require species-specific knowledge of vector life history and integrating such knowledge with avian ecology, including habitat requirements.

Although fragmentation can create opportunities for contact between hosts and vectors, it can also introduce barriers to movement. Many species of birds are less likely to move through open areas or gaps in the canopy relative to forested habitat, which can reduce functional connectivity for birds or even persistence in fragmented landscapes. Reduced connectivity can lead to population isolation and inbreeding, which has been linked with reduced immune function in birds including song sparrows (*Melospiza melodia*; Reid et al. 2007). Small remnant habitat patches may promote host contact if subpopulations become more spatially clustered. Reduced movement capacity from fragmentation may decrease opportunities for contact between hosts and between hosts and vectors across the landscape; however, hosts may also have fewer opportunities to leave habitat patches with high host or vector densities.

In addition to these effects on contact rates, habitat loss and fragmentation can impact pathogen dynamics through shifts in host and vector physiology. Habitat degradation associated with land use change can increase susceptibility to infection in hosts through interactions between physiological stress responses and immune function

(see also Chapter 3), though acute stressors such as those stimulated by predators can also enhance immune responses in some cases (Martin 2009). Habitat loss can be associated with increased stress due to the higher levels of disturbance or nutritional stress in altered landscapes. For example, nestling Eurasian treecreepers (*Certhia familiaris*) sampled from small forest patches had higher heterophil:lymphocyte ratios (a marker of stress) relative to nestlings in large patches (Suorsa et al. 2004). Moreover, their ratios were positively associated with mortality and poor body condition (Suorsa et al. 2004). Similarly, willow tits (*Poecile montanus*) in younger forests had higher baseline stress levels and decreased stress responses than tits in old growth forest patches, indicating chronic stress (Cīrule et al. 2017). While these studies point to potential impacts of land use change on host physiology and thus susceptibility, the immune or stress endpoints examined in these studies may not always be predictive of avian susceptibility to infection (see Chapter 3). Thus, further work is needed to understand how land use impacts host physiology in ways directly relevant for susceptibility to pathogens. Finally, habitat clearing and any subsequent increases in temperature can impact vector physiology, including greater vectorial capacity (Lefèvre et al. 2013) and more aggressive biting behavior (Vittor et al. 2006).

9.2.2 Chemical pollutants (toxicants)

Virtually all types of anthropogenic land use change are accompanied by increases in local concentrations of various pollutants in the air, water, and soil. Toxicants are toxic substances introduced to the environment or produced in artificially high quantities by industrial activities such as manufacturing or transportation. The exact types of pollutants that serve as toxicants depend on the presence of industrial activity, control of unwanted plants or animals, and other anthropogenic activities. For instance, urban areas can have higher concentrations of lead in nearby soil due to vehicles on roads (Qiao et al. 2014), which may explain parallel increases in lead concentrations in some urban birds (e.g., kestrel, *Falco tinnunculus*; Garcia-Fernandez et al. 2005). In agricultural areas, large quantities of pesticides are

applied to promote the growth of some species and inhibit others, and these products are associated with declines in aerial insectivorous birds (Spiller and Dettmers 2019). Although different pollutants enact their damaging effects on host physiology using different mechanisms, many interfere with metabolism and biochemical reactions, causing immunotoxic or immunosuppressive effects (Ferne et al. 2005; see also Chapter 3).

9.2.2.1 Heavy metals

Metals are deposited into the air, soil, and water through industrial activities such as mining, smelting, metallurgy, and vehicles on roads and the application of fertilizers, manures, and wastewater (He et al. 2005). Of these, heavy metals such as mercury (Hg), lead (Pb), cadmium (Cd), arsenic (As), chromium (Cr), and thallium (Tl) are harmful to wildlife and humans because they are toxic at low doses and cause organ damage, reduced reproductive success, and mortality in a wide variety of species (reviewed in Sánchez-Virosta et al. 2015). These elements are especially harmful because they are persistent in the environment, mobile, and able to bioaccumulate in tissues (Bauerová et al. 2017). Birds are often exposed to heavy metals through consuming contaminated food items such as fish and insects (Godwin et al. 2016). Likely due to this environmental and dietary exposure, the concentration of heavy metals is often higher in birds sampled in contaminated aquatic environments (Hollamby et al. 2004).

Increased exposure to heavy metals in contaminated landscapes can alter avian disease dynamics directly or indirectly. First, heavy metals can decrease host abundance, and thus contact rates between hosts and/or vectors, by causing direct mortality. Historically, the heavy metal most frequently associated with avian mortality has been lead and its distribution correlates to regions where hunting is legal and a common activity. Lead toxicity, through the ingestion of lead pellets used in shotgun shells, has (Pain 1996) and continues to kill thousands of waterfowl every year. Although mortalities have declined due to the 1985 US federal ban on the use of lead pellets for hunting waterfowl, lead pellets are still found in a wide range of waterfowl and predatory birds (Anderson et al.

2000). Lead toxicity is one of the main obstacles for the recovery of the California condor (*Gymnogyps californianus*; Kelly et al. 2014), which consume them by scavenging on remnants of game animals left by hunters. Likewise, birds of prey suffer significant mortality because they consume lead fragments contained in smaller prey, often left behind by hunters (e.g., prairie dogs [*Cynomys ludovicianus*] hunted as pest control; Stephens et al. 2005). Finally, population-level effects from lead toxicity have been documented for aquatic species that swallow fishing gear (e.g., loons; Haig et al. 2014). Although this has not been explicitly documented in birds, in some cases direct mortality from heavy metals may preferentially occur for hosts that are parasitized, which may have the effect of reducing disease transmission by removing infectious hosts from a population.

Sublethal effects to birds from heavy metals can result in the form of an increase in host susceptibility to pathogens. In experimental settings, birds fed metal-contaminated feed exhibited suppressed antibody production, leukocyte proliferation, and humoral and cell-mediated immune responses (Whitney and Cristol 2017). In free-living birds, individuals with greater metal contamination in their feathers can have higher stress hormone levels (e.g., cadmium and lead in common blackbirds; Meillère et al. 2016) and higher heterophil:lymphocyte ratios (e.g., great tits [*Parus major*]; Bauerová et al. 2017), indicating chronic stress. This relationship can be complicated and scale-dependent. Tree swallows (*Tachycineta bicolor*) feeding at mercury-contaminated sites exhibited reduced immune responses, but this relationship was not correlated with individual mercury loads (Hawley et al. 2009). Similarly, great tits sampled closer to a smelter exhibited lower humoral immune responses (Snoeijs et al. 2004). While exposure to lead, cadmium, and arsenic reduces host immune responses, Vermeulen et al. (2015) found that there was no population-level impact. Still other studies find no effect of heavy metal exposure on immunocompetence (Baos et al. 2006) and some find that metal exposure stimulates immune activity (Eeva et al. 2005). This variation may be at least partially explained by the timing of exposure as heavy metal exposure is more damaging to immature birds than adults.

The consequences of these pollutant-driven physiological shifts for host susceptibility to infection are not well studied. In zebra finches (*Taeniopygia guttata*), experimental exposure to methylmercury did not significantly alter coccidian parasite load or clearance rate, although for a short period during the course of infection, methylmercury-exposed birds had slightly higher coccidian counts (Ebers Smith et al. 2018). In France, house sparrows in highly urban habitats had higher lead concentrations in their feathers and were more likely to be infected with a malarial parasite, *Plasmodium relictum*, although in that study they did not find a clear relationship between lead concentrations, infection risk, and body condition (Bichet et al. 2013). Given the variability in avian host responses to metal exposure and the levels of contamination in disturbed areas, this is a rich area for future study.

9.2.2.2 Persistent organic pollutants

Persistent organic pollutants (POPs) are organic compounds typically used as pesticides (e.g., organochlorine pesticides, OCPs), industrial chemicals (e.g., polychlorinated biphenyls, PCBs), and pharmaceuticals that are resistant to degradation in the environment (Jones and de Voogt 1999). One of the most infamous POPs is dichlorodiphenyltrichloroethane (DDT), the negative effects of which are well known for birds, most notably decreased reproductive success through eggshell thinning (Bitman et al. 1970). Other POP pesticides, including heptachlor and diazinon, have been implicated in large bird 'die-off' events, likely through acute toxicity (Mittra et al. 2011). POPs are also able to bioaccumulate in fatty animal tissues, rendering them more harmful to carnivorous birds such as urban Cooper's hawks (*Accipiter cooperii*; Brogan et al. 2017).

In addition to reduced reproductive success and mortality, which directly reduce host abundance, chronic exposure to POPs can alter disease dynamics by altering host immunity and thus susceptibility to parasites or pathogens. Organochlorine compounds such as PCBs have demonstrated immunotoxic effects in several wildlife species. For example, glaucous gulls (*Larus hyperboreus*) can exhibit impaired humoral and cell-mediated immune

responses (Bustnes et al. 2004) and higher parasite loads (Sagerup et al. 2000) following PCB exposure. Because these compounds bioaccumulate in tissues, it is expected that these health effects will be more severe in predatory species at higher trophic levels.

While the application of pesticides in urban and agricultural lands will likely increase disease risks for birds by increasing susceptibility to infection, pesticides may also decrease the likelihood of encountering pathogen vectors. The impacts of pesticides on insect communities can be profound, reducing biomass by half in some contexts (Rioux et al. 2013). A decrease in vector abundance or diversity may explain why blood parasite prevalence is lower in urban birds (Santiago-Alarcon et al. 2020) and blood parasite richness is lower in disturbed (i.e., agricultural and urban) landscapes (Hernández-Lara et al. 2020). Studies that integrate vector abundance, host abundance, and host susceptibility in altered landscapes will be critical in disentangling these complex and sometimes opposing effects of anthropogenic land use change.

9.2.2.3 Other chemical pollutants

Second-generation anticoagulant rodenticides (SGARS) are potent rodenticides with long half-lives that increasingly appear to present a significant risk to wildlife. The US Environmental Protection Agency (EPA) determined that SGARS present a significant risk to wildlife (Erickson and Urban 2004), leading to the ban of most SGARS for consumer use, although use by pest managers and agricultural managers is still allowed. Thus toxicity from SGARS is associated with areas where pest control is needed and heavily managed. Primary mortality from consumption of prey affected by rodenticides has been recognized for some time. However, through improvement in diagnostic techniques and the recognition of sublethal lesions, secondary rodenticide toxicity is now seen as a significant health problem for birds of prey, especially species whose diet contains a large proportion of pest rodents (Murray 2017). In at least one study, rodenticide exposure has been associated with infection risk. In Spain, liver concentrations of rodenticides chlorophacinone and flocoumafen were positively correlated with parasite richness (i.e., number of parasite species) in great bustards (*Otis*

tarda; Lemus et al. 2011). Further study is needed to assess the impacts of rodenticide exposure for avian immune function and susceptibility to infection.

The impact of plastics on birds from consumption and entanglement cannot be ignored. Although population-level effects have not been clearly elucidated, ongoing research may bring to light the potential for plastics to be one of the most significant causes of direct and indirect mortality for all organisms, including birds. The consumption of plastic by birds has been documented since the early 1960s; however, only recently has the frequency of mortality events, particularly affecting seabirds, come to light (Wilcox et al. 2015). In addition to large particles that cause inflammation, malabsorption, and blockage of the gastrointestinal tract, microplastics have been found in seawater and in fish and various other items consumed by birds (Savoca et al. 2016). Further, microplastics can bind to other pollutants, such as heavy metals and PCBs, and thus their consumption can have an additive effect by leading to a higher consumption of pollutants (Jinhui et al. 2019). Lastly, compounds leaching from plastics themselves (e.g., OCPs and PCBs) have been known to cause a variety of health effects for birds (Colabuono et al. 2010; Teuten et al. 2009). Due to poor waste management systems, plastic products can end up in regions far from their production and use (Eriksen et al. 2014). The impact of plastics on pathogen dynamics is an area ripe for future study in birds and other wildlife.

9.2.3 Light and noise pollution

Although not a pollutant in the traditional sense, light pollution is gaining attention for its pervasive effects on ecosystem function (Navara and Nelson 2007). Light pollution, otherwise known as artificial light at night, occurs when light levels at night exceed 10% of natural illumination from lighting roads, homes, commercial buildings, and vehicles (Le Tallec 2019). Light pollution is now considered to be the fastest growing type of pollution on Earth, increasing on average by 6% of its global extent per year (Hölker et al. 2010).

Wildlife exposed to light pollution undergo a variety of behavioral and physiological shifts in the timing of activity, reproduction, and migration

(Ouyang et al. 2017). Although relatively much less is known about the impacts of light pollution on disease ecology, it is plausible that such conditions could impact avian susceptibility to infection. Artificial light at night has been associated with higher baseline corticosterone in great tits (Ouyang et al. 2015), which may decrease host immune function if such stressors are long term. Few studies to date have linked artificial light at night with infection risk in birds. Kernbach et al. (2019) experimentally exposed house sparrows to artificial light at night to test its effects on West Nile virus (WNV) infections. Sparrows exposed to artificial light for most of the night were on average infectious for two days longer than control birds, which the authors estimated could increase the risk of a WNV outbreak by an estimated 41% (Kernbach et al. 2019). In the same study, birds exposed to artificial light at night did not appear to have altered stress responses; however, immune responses appeared to be dysregulated based on changes in gene expression. In a related study, also on house sparrows, artificial light at night suppressed melatonin levels and increased WNV-induced mortality (Kernbach et al. 2020). In addition to physiological changes in hosts, artificial light at night may change the timing or frequency of interactions between hosts and vectors (Kernbach et al. 2018). For example, the crepuscular periods when both mosquitoes and sparrows are active may be extended in illuminated areas.

Noise pollution, or the elevation of background noise by anthropogenic activities such as vehicular traffic, also has diverse demonstrated effects on birds. In terms of effects potentially relevant to disease dynamics, studies have shown that birds exposed to traffic noise have elevated stress levels (Injaian et al. 2018). Few studies have examined the effect of traffic noise on immune function and what has been done shows no clear relationship (Crino et al. 2013). Nevertheless, if traffic noise promotes chronic stress in birds, it may ultimately increase avian susceptibility to infection over time.

9.2.4 Anthropogenic resources

A natural consequence of land transformation, including urbanization, is the creation and concentration of resources, such as water and food, that

attract birds. These resource sites for wild birds can be either intentional (e.g., bird feeders, zoos, parks where birds are fed) or unintentional (e.g., water features, landfills, wastewater treatment sites, constructed wetlands). Bird feeding is a multi-billion dollar industry in the US, where approximately 60 million households purchase bird feed and related items annually (US Department of the Interior et al. 2018). Common to these sites is the concentration of large numbers of birds of various species comingling in small areas, which can have specific implications for contact rates within and between host species, nutrition, and physical condition (Becker et al. 2015; Murray et al. 2016). Some types of anthropogenic resources also concentrate pathogen loads on inanimate surfaces ('fomites') that hosts contact and/or facilitate longer pathogen survival in the external environment, thus increasing the rate at which hosts contact pathogens in the environment (see also Chapter 4).

9.2.4.1 Supplemental feeding

Bird feeders are known to facilitate pathogen transmission for what are commonly known as 'backyard birds' (Passeriformes, Columbiformes, Piciformes, etc.). Evidence exists for bird feeders facilitating the transmission of the pathogens that cause at least five notorious 'feeder-associated' diseases: mycoplasmosis, aspergillosis, trichomoniasis, salmonellosis, and avian pox (Adelman et al. 2015); however, others such as avian mange and other types of infestations with macroparasites are likely to also have associations with bird feeders. For these diseases, the spatiotemporal overlap between feeding activities and outbreaks and the specific characteristics of the etiologic agent (e.g., the environmental persistence of *Salmonella*) have lent evidence to this association (Altizer et al. 2004; Lawson et al. 2018; Tizard 2004). For example, avian salmonellosis outbreaks tend to occur in the winter, when bird feeding activities peak, and the persistence of *Salmonella* on avian feces on feeders, particularly those that allow for defecation on food items, has been well demonstrated (Lawson et al. 2014). Unfortunately, pathogen transmission at bird feeders can also have public health significance (see also Chapter 12), as *Salmonella* strains circulating among passerines during outbreaks in both the UK and

US match genotypes also known to cause human salmonellosis

Feeders can impact pathogen dynamics in birds in myriad ways. In addition to behaviorally concentrating hosts, which augments contact rates within and among bird species, feeders modulate one of the major drivers of population regulation—food. Thus, supplemental resources from feeders can result in a higher abundance of avian hosts (Plummer et al. 2019), potentially increasing the magnitude of infectious disease outbreaks. Bird feeders and water baths, another form of supplemental resource for wildlife, also serve as environmental fomites for the persistence and spread of many types of pathogens, including fecal–oral (e.g., *Salmonella*), those spread by salivary or crop secretions (e.g., *Trichomonas gallinae*; Purple et al. 2015), and those spread from conjunctival secretions (i.e., *Mycoplasma gallisepticum*; Dhondt et al. 2007; Moyers et al. 2018).

Feeders may also indirectly influence disease dynamics by altering avian community composition. A recent study in the UK found that bird feeders were associated with increasing bird diversity and community evenness over time, which the authors attribute to the diversification of bird food types offered in backyards (Plummer et al. 2019). In New Zealand, Galbraith et al. (2015) found that supplemental feeding heavily impacted avian assemblages, with an increase in introduced species and a negative effect on a native insectivore. Changes in community structure due to bird feeders may have important consequences for pathogen transmission, particularly for multihost pathogens, because some species are more competent reservoirs than others. For example, in North America, pine siskins (*Spinus pinus*) have been suspected to be important in the epidemiology of avian salmonellosis because they form large flocks and are very aggressive at feeders (Daoust and Prescott 2007; Hernandez et al. 2012). The ability of feeders to amplify or dilute pathogen transmission by altering avian community composition is an important area for future work.

Although feeders are generally predicted to augment disease outbreaks through the above mechanisms, supplemental feeding can also potentially dampen disease outbreaks by augmenting host immunocompetence through improved nutritional

condition (Becker et al. 2018). Although nutritional status has long been associated with immunocompetence in birds (Klasing 1998; see also Chapter 3), the relationships among backyard feeders, nutritional status, and infection prevalence are likely complex and species-specific (Galbraith et al. 2017). For example, Wilcoxon et al. (2015) manipulated bird feeder presence at forested sites and found that birds at feeder-supplemented sites had higher body condition but also higher prevalence of clinical signs of parasite and pathogen infection. Thus, while feeders may augment nutritional status for some species, the increased contact rates associated with feeders may overwhelm any nutritional benefits. Additionally, commercial bird seed diets are likely deficient in important micronutrients and overdependence might negatively impact immunity, although this has not been specifically tested.

The role of individual heterogeneity in the use of bird feeders and the consequences of this variation for disease transmission are areas ripe for further study. Adelman et al. (2015) found that house finches (*Haemorrhous mexicanus*) that spend the longest amount of time on feeders per day are more likely to both acquire and transmit *M. gallisepticum*. In fact, behavioral plasticity is a characteristic of birds that successfully live near humans, such as those that take advantage of backyard feeders. Yet innovative foraging strategies are associated with higher contact with pathogens (Audet et al. 2016). Future work should examine how heterogeneity within and among avian species in feeder use impacts pathogen dynamics via diverse mechanisms. Although there is growing evidence that backyard bird feeders augment pathogen risk for a suite of bird species, the social impact it has on people and their connection with nature may be significant enough to discourage restrictive management (Cox and Gaston 2015).

9.2.4.2 Unintentional anthropogenic resources

The role of unintentional anthropogenic resources, such as water features, landfills, wastewater treatment sites, animal feedlots, and constructed wetlands, in driving avian disease dynamics is relatively less studied than the role of backyard bird feeding. Nonetheless, unintentional resources likely have very similar effects to those of backyard bird

feeders by augmenting contact between individuals within and among species. Here, we briefly highlight two characteristics that are unique to some forms of unintentional anthropogenic resources. First, sites like animal feedlots can bring wild birds into close proximity with agricultural animals, increasing the risk of transmission of pathogens across the agricultural–wild bird interface (see also Chapter 11). For example, sites like dairy farms are preferred foraging sites for American crows (*Corvus brachyrhynchos*), and foraging on highly concentrated resources likely puts them at risk of acquiring *Campylobacter jejuni* via fecal–oral transmission (Taff et al. 2016). Notably, American crows carry strains of *C. jejuni* that are similar to human and livestock strains (Weis et al. 2014), suggesting the potential for transmission across the wild bird–human–agricultural interface (see also Chapter 12).

Unintentional anthropogenic resources such as wastewater treatment sites also have the potential to augment pathogen persistence and virulence. For example, *Clostridium botulinum* toxicity is common in wastewater treatment plants because key environmental factors for toxin production (high temperatures, low oxygen tension, and decay of organic material) are inherent in these systems. Similarly, in the southeast US, mortality of wading birds, particularly of juveniles, from high loads of the intestinal nematode *Eustrongylides ignotus* is known to occur in sites with high nutrient inputs and is considered a risk for birds using wastewater ponds (Coyner et al. 2002). Antimicrobial resistance development in wastewater treatment ponds and the role that birds play in their carriage and dissemination is also a recently recognized concern (Martín-Maldonado et al. 2020). Although unlikely to cause outright mortality, much like *C. jejuni* transmission, it provides a One Health framework with which to examine how avian and public health are rooted in the health of the environment (see Section 9.4).

9.3 Overall consequences of land use change for avian parasitism: urbanization as a case study

Identifying generalizable relationships between land use change and avian disease dynamics is

important for predicting the impacts of future development for avian conservation, community ecology, and zoonotic disease risk. As may be evident from the previous sections, characterizing such an overall relationship is difficult due to the myriad effects of land use change that simultaneously affect contact rates between birds, contact between birds and vectors, and infection competence and susceptibility. Previous reviews have examined the impacts of land use change on wildlife disease prevalence across many taxa (Brearley et al. 2013; Gottdenker et al. 2014) and for birds in specific habitat types (Martin and Boruta 2014; Sehgal 2010; Sehgal 2015). Such efforts have found marked variation in the impacts of land use change on wildlife health depending on the type of land use change and mechanism of pathogen transmission. For example, Brearley et al. (2013) reported that just over half (53%) of wildlife studies detected an increase in infection prevalence due to human-associated land use change, especially in urban landscapes, while Chung et al. (2018) found that proportionately more bacterial pathogens had been reported in birds sampled in undisturbed landscapes relative to agricultural and urbanizing areas. Such variation is expected given the many mechanisms that may alter disease dynamics following changes in land use (Figure 9.1). To reconcile contrasting results between studies, mechanistic approaches that combine changes in communities (e.g., community composition) and individuals (e.g., susceptibility) with relevant abiotic factors (e.g., temperature gradients) could help identify the most salient changes that promote infection. Meta-analyses could also help identify more generalizable patterns to overcome site-specific effects.

There has been a recent surge in interest in the effects of urbanization, in particular on avian disease ecology, due to the rapid increase in urban areas around the globe. Much like the broader studies of land use changes noted above, studies of urbanization and avian health have found positive, negative, or no detectable relationships between urbanization and parasite prevalence, potentially due to variation in urban landscapes and host, vector, and pathogen life histories. Parasite prevalence in some systems is higher for urban birds. For

example, urban house finches had a higher prevalence of poxvirus infections and more severe coccidial infections relative to conspecifics in less urban habitats (Giraudeau et al. 2014). In other systems, urban birds fare better than their rural counterparts (e.g., Eurasian blackbirds and *Haemoproteus* spp.; Geue and Partecke 2008). In several studies, the relationships between urbanization and disease risk are complex. For example, urban blackbirds (*Turdus merula*) in Poland were less likely to carry ticks, but ticks in urban areas were more likely to be infected with *Borrelia* spp. (Gryczyńska 2018). For perhaps the quintessential urban bird, the pigeon or rock dove (*Columba livia*), the prevalence of haemosporidian parasites (*Haemoproteus* spp. and *Plasmodium* spp.) increased with urbanization for paler but not darker morphotypes (Jacquin et al. 2013). Because urbanization can affect a variety of biotic and abiotic conditions that impact host ecology, host physiology, and vector ecology, synthetic approaches to urban avian disease ecology are helpful in disentangling this complexity.

Given the diverse sources of variation underlying these host–pathogen relationships, a meta-analysis was conducted to identify any generalizable relationships between urbanization and wildlife health (Murray et al. 2019). This analysis included 106 studies found through a systematic search that compared urban and non-urban populations of the same species in terms of four measures of health: stress biomarkers, body condition, toxicant concentrations, and parasite prevalence or intensity (see Murray et al. 2019 for details on metrics). Of these, 46 studies focused on birds and included 159 different health comparisons (Figure 9.2). We analyzed this subset focused on birds using mixed effects models that included the study's standardized effect size (r) as the response variable (with positive values representing a positive health effect), health metric as a fixed effect, and study as a random effect. Urban birds had significantly higher toxicant concentrations than their rural counterparts, as reflected by negative effect sizes (Figure 9.2A), but there were no statistically significant differences in the other examined traits, including body condition, stress biomarkers, or parasitism (either prevalence or intensity) (for details on methods and these results see Murray et al. 2019).

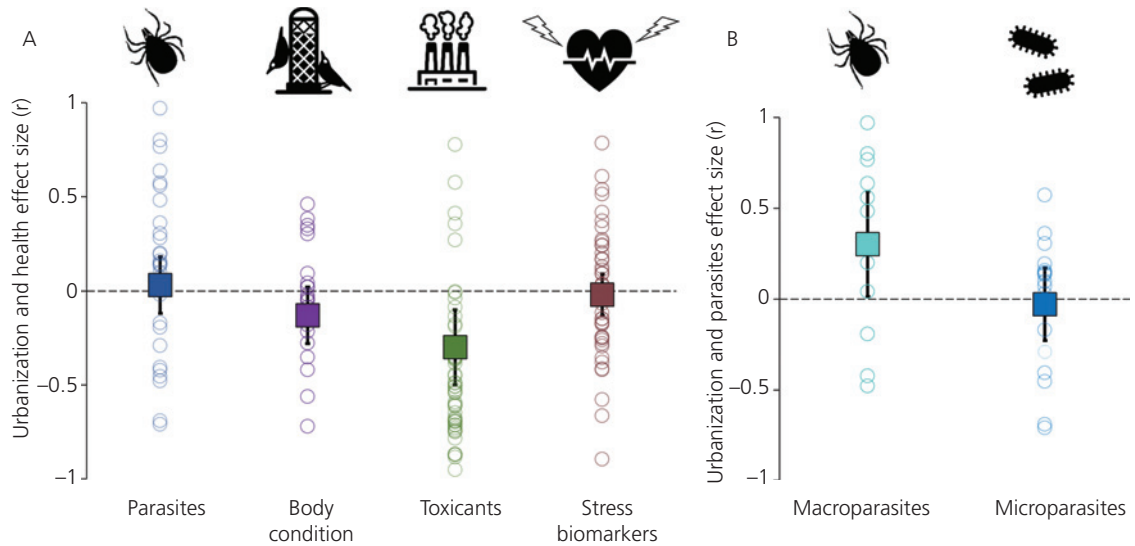


Figure 9.2 Living in urban areas might affect several metrics of avian health in positive (i.e., higher r values) or negative (i.e., lower r values) ways. (A) Relationships between urbanization and four metrics of health from 46 studies comparing the health of urban and non-urban bird populations (see Murray et al. 2019 for full results and explicit definitions of health metrics used for each category). Squares represent predicted means and 95% confidence intervals from mixed-effects models. (B) The 26 studies that quantified parasite prevalence in urban and non-urban populations are divided into macroparasites (e.g., arthropods, helminths) and microparasites (e.g., bacteria, viruses; see Chapter 2 for more on micro- and macroparasites). For both graphs, a positive value of r indicates a positive health effect (i.e., lower parasite prevalence, higher body condition, lower toxicant load, lower stress biomarkers) associated with urbanization. (Icons were created by Stewart Lamb Crowar, Maxim Kulikov, Iconographer, and Symbolon from Noun Project.)

To determine whether effects of urbanization emerge only for certain parasite types, we examined the subset of avian studies that quantified differences in parasite prevalence ($n = 26$) to determine whether urban versus rural birds differ in the prevalence of microparasites (bacteria: $n = 1$ studies, viruses: $n = 3$, protozoa: $n = 14$) versus macroparasites (helminths: $n = 3$, arthropods: $n = 9$). Within this subset, urban birds were more likely to have a lower prevalence of macroparasite infections relative to rural birds (i.e., a positive health effect and positive r value), but we detected no statistically significant differences in microparasite prevalence between the two groups (Figure 9.2B, $n = 26$; macroparasites: $\beta = 0.31 \pm 0.14$, $p = 0.05$; microparasites: $\beta = -0.37 \pm 0.17$, $p = 0.15$). This is consistent with the hypothesis that urban birds may benefit from the anthropogenic resources associated with urbanization in some way—perhaps because they enable hosts more time to remove macroparasitic arthropods (i.e., ectoparasites) via grooming (Murray et al. 2018). Thus, though urban birds may

experience negative health effects via increased exposure to toxicants, these potential negative health consequences are not reflected in parasite prevalence, which instead was lower for some types of parasites for urban birds. The relationships between urbanization and avian disease are complex and require more understanding of the underlying mechanisms associated with urban land use change.

Part of the observed variation in responses to urbanization likely stems from shifts in abiotic conditions and their impacts on vector populations. For instance, the heat island effect (see Section 9.1.1) may increase the risk of vector-borne pathogens for birds via mosquito abundance and biting behavior (Vittor et al. 2006). Novel microhabitats in urban environments can also promote standing water and provide breeding habitat for *Aedes aegypti* and *Culex quinquefasciatus*. These changes in vector ecology may be driving observed increases in avian blood parasites, such as West Nile virus (e.g., Bradley et al. 2008), in urban areas. Other vectors can be negatively impacted by urbanization if they rely on

particular vegetation types to find hosts, for example as occurs for *Ixodes* ticks in unmanaged grassy areas. Blackbirds sampled in urban sites in Germany were less likely to be infested with *Ixodes* ticks than in non-urban forests, potentially because urban habitats are less suitable for tick populations (Geue and Partecke 2008). Greater collaboration between ornithologists and entomologists will help identify such links between vector ecology and avian disease prevalence in urban areas.

9.4 Synthesis: need for multidisciplinary approaches

To understand the broader consequences of land use change for avian pathogen prevalence, it can be useful to apply the concept of One Health which encourages viewing health challenges through a multidisciplinary lens that emphasizes the commonalities among human, animal, and environmental health. In contexts in which land use change promotes higher pathogen prevalence in avian populations, there will likely be several impacts on other sectors, such as public, domestic, and environmental health. These effects are most notable for multi-host pathogens that, in addition to wild birds, can infect other wildlife, domestic animals, and/or humans (see also Chapters 11 and 12). With an increase in infected avian hosts, the prevalence of infected vectors (for vector-borne systems) will likely also increase, potentially leading to a higher infection prevalence in any non-avian hosts for a given system. For pathogens that are persistent in the environment (e.g., *Salmonella*), shifts in land use that promote infection in birds may lead to a higher environmental pathogen pool through fecal contamination, resulting in potential health risks for both domestic animals and humans. Additionally, birds can act as important and tractable means of monitoring other environmental sources of *Salmonella* spp. For example, a high diversity of *Salmonella* genotypes isolated from white ibis (*Eudocimus albus*) in urban areas indicates that they are reflecting environmental sources, which also infect people (Hernandez et al. 2016).

This increased shedding of pathogens in local environments can act in concert with increased opportunities for direct contact between wild birds

and domestic animals in urban and agricultural settings, reinforcing positive feedback cycles between land use change and pathogen dynamics that favor avian disease outbreaks. Furthermore, if high infection prevalence leads to high mortality rates for birds (e.g., from botulism), these processes may lead to disruptions in ecosystem services provided by wild birds such as seed dispersal, pollination, and consuming pests and pathogen vectors. All of these processes (vector infection rates, environmental contamination, domestic animal health, ecosystem services) impact public health through the transmission of zoonotic pathogens or environmental health. Thus, the health of birds in changing landscapes is intimately connected to the health of other wildlife, domestic animals, and humans and is fundamentally a One Health issue (see Chapter 13).

A growing body of work has documented the impacts of land use change for avian disease ecology in varied contexts, but the complexity and variability of study results highlight areas for future research. As previously mentioned, more studies are needed that link abiotic and biotic factors accompanying land use change with specific epidemiological processes such as host contact or physiology (Figure 9.1). If studies are timed appropriately, land use changes can provide ‘natural’ experiments and opportunities to measure changes in prevalence or other relevant factors following a before–after control–impact design. An increased focus on particular mechanisms (e.g., temperature gradients) could also help tease apart the effects of land use change from global climate change (see Chapter 10). Importantly, land use changes do not act in isolation from other anthropogenic effects, and interactions between land use and climate-driven changes in disease dynamics are thus an important avenue for future research.

Given the variability in avian health with urbanization, systematic multicity comparisons are needed to examine the role of urban development and study location as contributing factors (Magle et al. 2019). Such studies will also benefit from greater knowledge about biologically relevant thresholds for toxicants in avian tissues arising from air and water pollutants. The effects of food provisioning are multifaceted, with potential

benefits (e.g., lower nutritional stress) and harmful effects (e.g., aggregation) on avian health. Field studies and experimental approaches can help identify the relative importance of diet quality, contact rates, and movement on pathogen exposure for urban birds. Studies addressing the mechanisms for the issues listed above will help predict changes in avian disease dynamics to advance bird conservation and public health. Finally, it is important to consider that here we have primarily discussed the consequences of land use change for parasite prevalence or intensity in birds; however, relatively few studies have directly linked parasite load to wild bird health or fitness (Delgado-V. and French 2012; see also Chapter 3). Ultimately, linking host-pathogen dynamics with avian survival and reproduction can inform management strategies to mitigate bird declines in changing landscapes.

Acknowledgments

We thank Cecilia A. Sánchez, Daniel J. Becker, Kaylee A. Byers, Katherine E.L. Worsley-Tonks, and Meggan E. Craft for their roles in collecting the data associated with the urban bird meta-analysis and Figure 9.2.

Literature cited

- Adelman, J.S., Moyers, S.C., Farine, D.R., and Hawley, D.M. (2015). Feeder use predicts both acquisition and transmission of a contagious pathogen in a North American songbird. *Proceedings of the Royal Society B: Biological Sciences*, 282, 20151429.
- Afrane, Y.A., Little, T.J., Lawson, B.W., Githeko, A.K., and Yan, G. (2008). Deforestation and vectorial capacity of *Anopheles gambiae* Giles mosquitoes in malaria transmission, Kenya. *Emerging Infectious Diseases*, 14(10), 1533–8.
- Altizer, S., Hochachka, W.M., and Dhondt, A.A. (2004). Seasonal dynamics of mycoplasmal conjunctivitis in eastern North American house finches. *Journal of Animal Ecology*, 73(2), 309–22.
- Anderson, W. L., Havera, S. P., and Zercher, B. W. (2000). Ingestion of lead and nontoxic shotgun pellets by ducks in the Mississippi Flyway. *Journal of Wildlife Management*, 64(3), 848–57.
- Antrop, M. and Van Eetvelde, V. (2000). Holistic aspects of suburban landscapes: visual image interpretation and landscape metrics. *Landscape and Urban Planning*, 50(1–3), 43–58.
- Atkinson, C.T. and LaPointe, D.A. (2009). Introduced avian diseases, climate change, and the future of Hawaiian honeycreepers. *Journal of Avian Medicine and Surgery*, 23(1), 53–63.
- Audet, J.-N., Ducatez, S., and Lefebvre, L. (2016). The town bird and the country bird: problem solving and immunocompetence vary with urbanization. *Behavioral Ecology*, 27(2), 637–44.
- Baos, R., Jovani, R., Forero, M.G., et al. (2006). Relationships between T-cell-mediated immune response and Pb, Zn, Cu, Cd, and As concentrations in blood of nestling white storks (*Ciconia ciconia*) and black kites (*Milvus migrans*) from Doñana (southwestern Spain) after the Aznalcóllar toxic. *Environmental Toxicology and Chemistry*, 25(4), 1153–9.
- Bauerová, P., Vinklerová, J., Hraníček, J., et al. (2017). Associations of urban environmental pollution with health-related physiological traits in a free-living bird species. *Science of the Total Environment*, 601–602, 1556–65.
- Becker, D.J., Streiker, D.G., and Altizer, S. (2015). Linking anthropogenic resources to wildlife–pathogen dynamics: a review and meta-analysis. *Ecology Letters*, 18(5), 483–95.
- Becker, D.J., Hall, R.J., Forbes, K.M., Plowright, R.K., and Altizer, S.M. (2018). Anthropogenic resource subsidies and host–parasite dynamics in wildlife. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1745).
- Bichet, C., Scheifler, R., Cœurdassier, M., Julliard, R., Sorci, G., and Loiseau, C. (2013). Urbanization, trace metal pollution, and malaria prevalence in the house sparrow. *PLoS One*, 8(1), e53866.
- Bitman, J., Cecil, H., and Fries, G. (1970). DDT-induced inhibition of avian shell gland carbonic anhydrase: a mechanism for thin eggshells. *Science*, 168(3931), 594–6.
- Boal, C.W., Mannan, W., and Hudelson, K.S. (1998). Trichomoniasis in Cooper's hawks from Arizona. *Journal of Wildlife Diseases*, 34(3), 590–3.
- Bradley, C.A. and Altizer, S. (2007). Urbanization and the ecology of wildlife diseases. *Trends in Ecology and Evolution*, 22(2), 95–102.
- Bradley, C.A., Gibbs, S.E.J., and Altizer, S. (2008). Urban land use predicts West Nile virus exposure in songbirds. *Ecological Applications*, 18(5), 1083–92.
- Brearely, G., Rhodes, J., Bradley, A., et al. (2013). Wildlife disease prevalence in human-modified landscapes. *Biological Reviews*, 88(2), 427–42.
- Brogan, J.M., Green, D.J., Maisonneuve, F., and Elliott, J.E. (2017). An assessment of exposure and effects of persistent organic pollutants in an urban Cooper's hawk (*Accipiter cooperii*) population. *Ecotoxicology*, 26(1), 32–45.
- Bustnes, J.O., Hanssen, S.A., Folstad, I., Erikstad, K.E., Hasselquist, D., and Skaare, J.U. (2004). Immune

- function and organochlorine pollutants in arctic breeding glaucous gulls. *Archives of Environmental Contamination and Toxicology*, 47(4), 530–41.
- Chasar, A., Loiseau, C., Valkiunas, G., Iezhova, T., Smith, T.B., and Sehgal, R.N.M. (2009). Prevalence and diversity patterns of avian blood parasites in degraded African rainforest habitats. *Molecular Ecology*, 18(19), 4121–33.
- Chung, D.M., Ferree, E., Simon, D.M., and Yeh, P.J. (2018). Patterns of bird–bacteria associations. *EcoHealth*, 15(3), 627–41.
- Cirule, D., Krama, T., Krams, R., et al. (2017). Habitat quality affects stress responses and survival in a bird wintering under extremely low ambient temperatures. *Die Naturwissenschaften*, 104(11–12), 99.
- Colabuono, F.I., Taniguchi, S., and Montone, R.C. (2010). Polychlorinated biphenyls and organochlorine pesticides in plastics ingested by seabirds. *Marine Pollution Bulletin*, 60(4), 630–4.
- Cox, D.T.C. and Gaston, K.J. (2015). Likeability of garden birds: Importance of species knowledge and richness in connecting people to nature. *PLoS ONE*, 10(11), 1–14.
- Coyner, D.F., Spalding, M.G., and Forrester, D.J. (2002). Epizootiology of *Eustrongylides ignotus* in Florida: transmission and development of larvae in intermediate hosts. *Journal of Wildlife Diseases*, 38(3), 483–99.
- Crino, O.L., Johnson, E.E., Blickley, J.L., Patricelli, G.L., and Breuner, C.W. (2013). Effects of experimentally elevated traffic noise on nestling white-crowned sparrow stress physiology, immune function and life history. *Journal of Experimental Biology*, 216(11), 2055–62.
- Daoust, P. and Prescott, J. (2007). Salmonellosis. In N. Thomas, D. Hunter, and C.T. Atkinson, eds. *Infectious Diseases of Wild Birds*, pp. 270–88. Blackwell Publishing, Oxford.
- Delgado-V., C.A. and French, K. (2012). Parasite–bird interactions in urban areas: current evidence and emerging questions. *Landscape and Urban Planning*, 105(1–2), 5–14.
- Dhondt, A.A., Dhondt, K.V., Hawley, D.M., and Jennelle, C.S. (2007). Experimental evidence for transmission of *Mycoplasma gallisepticum* in house finches by fomites. *Avian Pathology*, 36(3), 205–8.
- Dudley, N. and Alexander, S. (2017). Agriculture and biodiversity: a review. *Biodiversity*, 18, 45–9.
- Ebers Smith, J.H., Cristol, D.A., and Swaddle, J.P. (2018). Experimental infection and clearance of coccidian parasites in mercury-exposed zebra finches. *Bulletin of Environmental Contamination and Toxicology*, 100, 89–94.
- Eeva, T., Hasselquist, D., Langefors, Å., Tummeleht, L., Nikinmaa, M., and Ilmonen, P. (2005). Pollution related effects on immune function and stress in a free-living population of pied flycatcher *Ficedula hypoleuca*. *Journal of Avian Biology*, 36(5), 405–12.
- Ellis, E.C. and Ramankutty, N. (2008). Putting people in the map: anthropogenic biomes of the world. *Frontiers in Ecology and the Environment*, 6(8), 439–47.
- Erickson, W. and Urban, D. (2004). *Potential Risks of Nine Rodenticides to Birds and Nontarget Mammals: A Comparative Approach*. US EPA, Washington, DC.
- Eriksen, M., Lebreton, L.C.M., Carson, H.S., et al. (2014). Plastic pollution in the world's oceans: more than 5 trillion plastic pieces weighing over 250,000 tons afloat at sea. *PLoS ONE*, 9(12), e111913.
- Fahrig, L. (2003). Effects of habitat fragmentation on biodiversity. *Annual Review of Ecology, Evolution and Systematics*, 34, 487–515.
- Fernie, K.J., Mayne, G., Shutt, J.L., et al. (2005). Evidence of immunomodulation in nestling American kestrels (*Falco sparverius*) exposed to environmentally relevant PBDEs. *Environmental Pollution*, 138(3), 485–93.
- Gaertner, M., Wilson, J.R.U., Cadotte, M.W., MacIvor, J.S., Zenni, R.D., and Richardson, D.M. (2017). Non-native species in urban environments: patterns, processes, impacts and challenges. *Biological Invasions*, 19(12), 3461–9.
- Galbraith, J.A., Beggs, J.R., Jones, D.N., and Stanley, M.C. (2015). Supplementary feeding restructures urban bird communities. *Proceedings of the National Academy of Sciences of the United States of America*, 112(20), E2648–57.
- Galbraith, J. A., Stanley, M. C., Jones, D. N., and Beggs, J. R. (2017). Experimental feeding regime influences urban bird disease dynamics. *Journal of Avian Biology*, 48, 700–13.
- Garcia-Fernandez, A., Romero, D., Martinez-Lopez, E., Navas, I., Pulido, M., and Maria-Mojica, P. (2005). Environmental lead exposure in the European kestrel (*Falco tinnunculus*) from southeastern Spain: the influence of leaded gasoline regulations. *Bulletin of Environmental Contamination and Toxicology*, 74, 314–19.
- Geue, D. and Partecke, J. (2008). Reduced parasite infestation in urban Eurasian blackbirds (*Turdus merula*): a factor favoring urbanization? *Canadian Journal of Zoology*, 86, 1419–25.
- Giraudeau, M., Mousel, M., Earl, S., and McGraw, K. (2014). Parasites in the city: degree of urbanization predicts poxvirus and coccidian infections in house finches (*Haemorrhous mexicanus*). *PLoS ONE*, 9(2), e86747.
- Godwin, C.M., Smits, J.E.G., and Barclay, R.M.R. (2016). Metals and metalloids in nestling tree swallows and their dietary items near oilsands mine operations in northern Alberta. *Science of the Total Environment*, 562, 714–23.
- Gottdenker, N.L., Streicker, D.G., Faust, C.L., and Carroll, C.R. (2014). Anthropogenic land use change and infectious diseases: a review of the evidence. *EcoHealth*, 11, 619–32.

- Gryczyńska, A. (2018). Urban and forest-living blackbirds *Turdus merula* as hosts of *Borrelia* spp. infected ticks. *Polish Journal of Ecology*, 66(3), 309–14.
- Haig, S.M., D'Elia, J., Eagles-Smith, C., et al. (2014). The persistent problem of lead poisoning in birds from ammunition and fishing tackle. *The Condor*, 116(3), 408–28.
- Hamilton, A.J. (2007). Potential microbial and chemical hazards to waterbirds at the Western Treatment Plant. *Ecological Management and Restoration*, 8(1), 38–41.
- Hawley, D.M., Hallinger, K.K., and Cristol, D.A. (2009). Compromised immune competence in free-living tree swallows exposed to mercury. *Ecotoxicology*, 18, 499–503.
- He, Z.L., Yang, X.E., and Stoffella, P.J. (2005). Trace elements in agroecosystems and impacts on the environment. *Journal of Trace Elements in Medicine and Biology*, 19(2–3), 125–40.
- Hernandez, S.M., Keel, K., Sanchez, S., et al. (2012). Epidemiology of a *Salmonella enterica* subsp. *enterica* serovar *typhimurium* strain associated with a songbird outbreak. *Applied and Environmental Microbiology*, 78(20), 7290–8.
- Hernandez, S.M., Welch, C.N., Peters, V.E., et al. (2016). Urbanized white ibises (*Eudocimus albus*) as carriers of *Salmonella enterica* of significance to public health and wildlife. *PLoS ONE*, 11(10), 1–22.
- Hernández-Lara, C., Carbó-Ramírez, P., and Santiago-Alarcon, D. (2020). Effects of land use change (rural–urban) on the diversity and epizootiological parameters of avian *Haemosporidia* in a widespread neotropical bird. *Acta Tropica*, 209, 105542.
- Hölker, F., Moss, T., Griefahn, B., et al. (2010). The dark side of light: a transdisciplinary research agenda for light pollution policy. *Ecology and Society*, 15(4), 13–24.
- Hollamby, S., Afema-azikuru, J., Sikarskie, J.G., et al. (2004). Mercury and persistent organic pollutant concentrations in African fish eagles, Marabou storks, and Nile tilapia in Uganda. *Journal of Wildlife Diseases*, 40(3), 501–14.
- Injaian, A.S., Taff, C.C., Pearson, K.L., Gin, M.M.Y., Patricelli, G.L., and Vitousek, M.N. (2018). Effects of experimental chronic traffic noise exposure on adult and nestling corticosterone levels, and nestling body condition in a free-living bird. *Hormones and Behavior*, 106(July), 19–27.
- Jacquín, L., Récapet, C., Prévot-Julliard, A.C., et al. (2013). A potential role for parasites in the maintenance of color polymorphism in urban birds. *Oecologia*, 173(3), 1089–99.
- Jinhui, S., Sudong, X., Yan, N., Xia, P., Jiahao, Q., and Yongjian, X. (2019). Effects of microplastics and attached heavy metals on growth, immunity, and heavy metal accumulation in the yellow seahorse, *Hippocampus kuda* Bleeker. *Marine Pollution Bulletin*, 149(August), 110510.
- Jones, K. and de Voogt, P. (1999). Persistent organic pollutants (POPs): state of the science. *Environmental Pollution*, 100, 209–21.
- Kelly, T.R., Grantham, J., George, D., et al. (2014). Spatiotemporal patterns and risk factors for lead exposure in endangered California condors during 15 years of reintroduction. *Conservation Biology*, 28(6), 1721–30.
- Kernbach, M.E., Hall, R.J., Burkett-Cadena, N.D., Unnasch, T.R., and Martin, L.B. (2018). Dim light at night: physiological effects and ecological consequences for infectious disease. *Integrative and Comparative Biology*, 58(5), 995–1007.
- Kernbach, M.E., Newhouse, D.J., Miller, J.M., et al. (2019). Light pollution increases West Nile virus competence of a ubiquitous passerine reservoir species. *Proceedings of the Royal Society B: Biological Sciences*, 286(1907), 20191051.
- Kernbach, M.E., Cassone, V.M., Unnasch, T.R., and Martin, L.B. (2020). Broad-spectrum light pollution suppresses melatonin and increases West Nile virus-induced mortality in house sparrows (*Passer domesticus*). *The Condor*, 122, 1–13.
- Klasing, K.C. (1998). Nutritional modulation of resistance to infectious diseases. *Poultry Science*, 77(8), 1119–25.
- Lawson, B., De Pinna, E., Horton, R.A., et al. (2014). Epidemiological evidence that garden birds are a source of human salmonellosis in England and Wales. *PLoS ONE*, 9(2), 1–10.
- Lawson, B., Robinson, R.A., Toms, M.P., Risely, K., Macdonald, S., and Cunningham, A.A. (2018). Health hazards to wild birds and risk factors associated with anthropogenic food provisioning. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1745).
- Le Tallec, T. (2019). *What is the ecological impact of light pollution?* Encyclopedia of the Environment, <https://www.encyclopedie-environnement.org/en/life/what-is-the-ecological-impact-of-light-pollution/>.
- Lefèvre, T., Vantaux, A., Dabiré, K.R., Mouline, K., and Cohuet, A. (2013). Non-genetic determinants of mosquito competence for malaria parasites. *PLoS Pathogens*, 9(6).
- Lemus, J.A., Bravo, C., García-Montijano, M., et al. (2011). Side effects of rodent control on non-target species: rodenticides increase parasite and pathogen burden in great bustards. *Science of the Total Environment*, 409(22), 4729–34.
- Magle, S.B., Fidino, M., Lehrer, E.W., et al. (2019). Advancing urban wildlife research through a multi-city collaboration. *Frontiers in Ecology and the Environment*, 17(4), 232–9.

- Martin, L.B. (2009). Stress and immunity in wild vertebrates: timing is everything. *General and Comparative Endocrinology*, 163(1–2), 70–6.
- Martin, L.B. and Boruta, M. (2014). The impacts of urbanization on avian disease transmission and emergence. In D. Gil and H. Brumm, eds. *Avian Urban Ecology*, Issue I, pp. 116–28. Oxford University Press, Oxford.
- Martín-Maldonado, B., Vega, S., Mencía-Gutiérrez, A., et al. (2020). Urban birds: an important source of antimicrobial resistant *Salmonella* strains in central Spain. *Comparative Immunology, Microbiology and Infectious Diseases*, 72, 101519.
- Meillère, A., Brischoux, F., Bustamante, P., et al. (2016). Corticosterone levels in relation to trace element contamination along an urbanization gradient in the common blackbird (*Turdus merula*). *Science of the Total Environment*, 566–567, 93–101.
- Mendenhall, C.D., Archer, H.M., Brenes, F.O., Sekercioglu, C.H., and Sehgal, R.N.M. (2013). Balancing biodiversity with agriculture: land sharing mitigates avian malaria prevalence. *Conservation Letters*, 6(2), 125–31.
- Mitra, A., Chatterjee, C., and Mandal, F.B. (2011). Synthetic chemical pesticides and their effects on birds. *Research Journal of Environmental Toxicology*, 5(2), 81–96.
- Moyers, S.C., Adelman, J.S., Farine, D.R., Thomason, C.A., and Hawley, D.M. (2018). Feeder density enhances house finch disease transmission in experimental epidemics. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1745).
- Murphy, M.T. (2003). Avian population trends within the evolving agricultural landscape of eastern and central United States. *The Auk*, 120(1), 20–34.
- Murray, C.G., Kasel, S., Szantyr, E., Barratt, R., and Hamilton, A.J. (2014). Waterbird use of different treatment stages in waste-stabilisation pond systems. *Emu*, 114(1), 30–40.
- Murray, M. (2017). Anticoagulant rodenticide exposure and toxicosis in four species of birds of prey in Massachusetts, USA, 2012–2016, in relation to use of rodenticides by pest management professionals. *Ecotoxicology*, 26(8), 1041–50.
- Murray, M.H., Becker, D. J., Hall, R.J., and Hernandez, S.M. (2016). Wildlife health and supplemental feeding: a review and management recommendations. *Biological Conservation*, 204, 163–74.
- Murray, M.H., Kidd, A.D., Curry, S.E., et al. (2018). From wetland specialist to hand-fed generalist: shifts in diet and condition with provisioning for a recently urbanized wading bird. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373, 20170100.
- Murray, M.H., Sánchez, C., Becker, D.J., Byers, K.A., Worsley-Tonks, K.E.L., and Craft, M.E. (2019). City sicker? A meta-analysis of wildlife health and urbanization. *Frontiers in Ecology and the Environment*, 17(10), 575–83.
- Navara, K.J. and Nelson, R.J. (2007). The dark side of light at night: physiological, epidemiological, and ecological consequences. *Journal of Pineal Research*, 43(3), 215–24.
- Ogrzewalska, M., Uezu, A., Jenkins, C.N., and Labruna, M.B. (2011). Effect of forest fragmentation on tick infestations of birds and tick infection rates by *Rickettsia* in the Atlantic Forest of Brazil. *EcoHealth*, 8(3), 320–31.
- Oke, T.R. (1973). City size and the urban heat island. *Atmospheric Environment*, 7, 769–79.
- Ouyang, J.Q., De Jong, M., Hau, M., Visser, M.E., Van Grunsven, R.H.A., and Spoelstra, K. (2015). Stressful colours: corticosterone concentrations in a free-living songbird vary with the spectral composition of experimental illumination. *Biology Letters*, 11(8).
- Ouyang, J.Q., de Jong, M., van Grunsven, R.H.A., et al. (2017). Restless roosts: light pollution affects behavior, sleep, and physiology in a free-living songbird. *Global Change Biology*, 23(11), 4987–94.
- Owens, I.P.F. and Bennett, P.M. (2000). Ecological basis of extinction risk in birds: habitat loss versus human persecution and introduced predators. *Proceedings of the National Academy of Sciences of the United States of America*, 97(22), 12144–8.
- Pain, D.J. (1996). Lead in waterfowl. In *Environmental Contaminants in Wildlife: Interpreting Tissue Concentrations*, SETAC Special Publications Series, pp. 251–64. Lewis Publishers, Boca Raton, FL.
- Palacio, R.D., Kattan, G.H., and Pimm, S.L. (2019). Bird extirpations and community dynamics in an Andean cloud forest over 100 years of land-use change. *Conservation Biology*, 34(3), 677–87.
- Plummer, K.E., Risely, K., Toms, M.P., and Siriwardena, G.M. (2019). The composition of British bird communities is associated with long-term garden bird feeding. *Nature Communications*, 10(1).
- Purple, K.E., Humm, J.M., Kirby, R.B., Saidak, C.G., and Gerhold, R. (2015). *Trichomonas gallinae* persistence in four water treatments. *Journal of Wildlife Diseases*, 51(3), 739–42.
- Qiao, X., Schmidt, A.H., Tang, Y., Xu, Y., and Zhang, C. (2014). Demonstrating urban pollution using toxic metals of road dust and roadside soil in Chengdu, southwestern China. *Stochastic Environmental Research and Risk Assessment*, 28(4), 911–19.
- Reid, J.M., Arcese, P., Keller, L.F., Elliott, K.H., Sampson, L., and Hasselquist, D. (2007). Inbreeding effects on immune response in free-living song sparrows (*Melospiza melodia*). *Proceedings of the Royal Society B: Biological Sciences*, 274(1610), 697–706.
- Rioux, S., Garant, D., Pelletier, F., and Bélisle, M. (2013). Seasonal patterns in tree swallow prey (*Diptera*)

- abundance are affected by agricultural intensification. *Ecological Applications*, 23(1), 122–33.
- Ritchie, H. and Roser, M. (2020). *Land Use*. OurWorldInData. Org.
- Rosenberg, K.V., Dokter, A.M., Blancher, P.J., et al. (2019). Decline of the North American avifauna. *Science*, 366, 120–4.
- Sagerup, K., Henriksen, E.O., Skorpung, A., Skaare, J.U., and Gabrielsen, G.W. (2000). Intensity of parasitic nematodes increases with organochlorine levels in the glaucous gull. *Journal of Applied Ecology*, 37(3), 532–9.
- Sánchez-Virosta, P., Espín, S., García-Fernández, A.J., and Eeva, T. (2015). A review on exposure and effects of arsenic in passerine birds. *Science of the Total Environment*, 512–513, 506–25.
- Santiago-Alarcon, D., Carbó-Ramírez, P., Macgregor-Fors, I., Chávez-Zichinelli, C.A., and Yeh, P.J. (2020). The prevalence of avian haemosporidian parasites in an invasive bird is lower in urban than in non-urban environments. *Ibis*, 162, 201–14.
- Savoca, M.S., Wohlfeil, M.E., Ebeler, S.E., and Nevitt, G.A. (2016). Marine plastic debris emits a keystone infochemical for olfactory foraging seabirds. *Science Advances*, 2(11), 1–9.
- Sehgal, R. (2010). Deforestation and avian infectious diseases. *Journal of Experimental Biology*, 213(6), 955–60.
- Sehgal, R. (2015). Manifold habitat effects on the prevalence and diversity of avian blood parasites. *International Journal for Parasitology: Parasites and Wildlife*, 4(3), 421–30.
- Snoeijs, T., Dauwe, T., Pinxten, R., Vandesande, F., and Eens, M. (2004). Heavy metal exposure affects the humoral immune response in a free-living small songbird, the great tit (*Parus major*). *Archives of Environmental Contamination and Toxicology*, 46(3), 399–404.
- Spiller, K.J. and Dettmers, R. (2019). Evidence for multiple drivers of aerial insectivore declines in North America. *Ornithological Applications*, 121, 1–13.
- Stephens, R., Johnson, A., Plumb, R., Dickerson, K., McKinstry, M., and Anderson, S. (2005). Secondary lead poisoning in golden eagle and ferruginous hawk chicks consuming shot black-tailed prairie dogs, Thunder Basin National Grassland, Wyoming. Contaminants Report Number R6/720/05. *US Fish & Wildlife Publications*, 230.
- Suorsa, P., Helle, H., Koivunen, V., Huhta, E., Nikula, A., and Hakkarainen, H. (2004). Effects of forest patch size on physiological stress and immunocompetence in an area-sensitive passerine, the Eurasian treecreeper (*Certhia familiaris*): an experiment. *Proceedings of the Royal Society B: Biological Sciences*, 271(1537), 435–40.
- Taff, C.C., Weis, A.M., Wheeler, S., et al. (2016). Influence of host ecology and behavior on *Campylobacter jejuni* prevalence and environmental contamination risk in a synanthropic wild bird species. *Applied and Environmental Microbiology*, 82(15), 4811–20.
- Teuten, E.L., Saquing, J.M., Knappe, D.R.U., et al. (2009). Transport and release of chemicals from plastics to the environment and to wildlife. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 364(1526), 2027–45.
- Tizard, I. (2004). Salmonellosis in wild birds. *Seminars in Avian and Exotic Pet Medicine*, 13(2), 50–66.
- US Department of the Interior, US Fish and Wildlife Service, US Department of Commerce, and US Census Bureau (2018). *2016 National Survey of Fishing, Hunting, and Wildlife-Associated Recreation*. https://www.fws.gov/wsfrprograms/subpages/nationalsurvey/nat_survey2016.pdf.
- van Hoesel, W., Marzal, A., Magallanes, S., Santiago-Alarcon, D., Ibáñez-Bernal, S., and Renner, S.C. (2019). Management of ecosystems alters vector dynamics and haemosporidian infections. *Scientific Reports*, 9(1), 1–11.
- Venter, O., Sanderson, E.W., Magrath, A., et al. (2016). Sixteen years of change in the global terrestrial human footprint and implications for biodiversity conservation. *Nature Communications*, 7, 12558.
- Vermeulen, A., Müller, W., Matson, K.D., Tieleman, B.I., Bervoets, L., and Eens, M. (2015). Sources of variation in innate immunity in great tit nestlings living along a metal pollution gradient: an individual-based approach. *Science of the Total Environment*, 508(2015), 297–306.
- Vittor, A.Y., Gilman, R.H., Tielsch, J., et al. (2006). The effect of deforestation on the human-biting rate of *Anopheles darlingi*, the primary vector of falciparum malaria in the Peruvian Amazon. *American Journal of Tropical Medicine and Hygiene*, 74(1), 3–11.
- Weis, A.M., Miller, W.A., Byrne, B.A., et al. (2014). Prevalence and pathogenic potential of *Campylobacter* isolates from free-living, human-commensal American crows. *Applied and Environmental Microbiology*, 80(5), 1639–44.
- Whitney, M. and Cristol, D. (2017). Impacts of sublethal mercury exposure on birds: a detailed review. In *Reviews of Environmental Contamination and Toxicology*, pp. 113–63. Springer, Cham.
- Wilcox, C., Van Seville, E., Hardesty, B.D., and Estes, J.A. (2015). Threat of plastic pollution to seabirds is global, pervasive, and increasing. *Proceedings of the National Academy of Sciences of the United States of America*, 112(38), 11899–904.
- Wilcoxon, T.E., Horn, D.J., Hogan, B.M., et al. (2015). Effects of bird-feeding activities on the health of wild birds. *Conservation Physiology*, 3, 1–13.
- Zhong, D., Wang, X., Xu, T., et al. (2016). Effects of microclimate condition changes due to land use and land cover changes on the survivorship of malaria vectors in China–Myanmar border region. *PLoS ONE*, 11(5).

